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C:\Windows\System32>cd C:\Users\lohit\OneDrive\Documents\MongoDB\mongosh-2.10.0-win32-x64\bin

C:\Users\lohit\OneDrive\Documents\MongoDB\mongosh-2.10.0-win32-x64\bin>mongosh "mongodb+srv://cluster0.eay9vj1.mongodb.net/" --apiVersion 1 --username lohithsurya_db_user
Enter password: ****
Current Mongosh Log ID: 6a89317762f34d54205e2e26
Connecting to:
  mongodb+srv://<<credentials>>@cluster0.eay9vj1.mongodb.net/?appName=mongosh+2.10.0
MongoServerError: 0C310000:error:0A000438:SSL routines:ssl3_read_bytes:tlsv1 alert internal error:openssl\ssl\record\rec_layer_s3.c:918:SSL alert number 80
. It looks like this is a MongoDB Atlas cluster. Please ensure that your Network Access List allows connections from your IP.

C:\Users\lohit\OneDrive\Documents\MongoDB\mongosh-2.10.0-win32-x64\bin>mongosh "mongodb+srv://cluster0.eay9vj1.mongodb.net/" --apiVersion 1 --username lohithsurya_db_user
Enter password: ****
Current Mongosh Log ID: 6a89322969041f57525a779e
Connecting to:
  mongodb+srv://<<credentials>>@cluster0.eay9vj1.mongodb.net/?appName=mongosh+2.10.0
Using MongoDB:
  8.0.29 (API Version 1)
Using Mongosh:
  2.10.0

For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

Atlas atlas-4upy6x-shard-0 [primary] test> show databases;
BookFlowDB      8.00 KiB
sample_mflix    143.55 MiB
schoolDB         164.00 KiB
test            132.00 KiB
admin            0 B
local            0 B
Atlas atlas-4upy6x-shard-0 [primary] test> use BookFlowDB;
switched to db BookFlowDB
Atlas atlas-4upy6x-shard-0 [primary] BookFlowDB: db.createCollection("book_metadata")
{ ok: 1 }
Atlas atlas-4upy6x-shard-0 [primary] BookFlowDB: db.book_metadata.insertMany([
{
  title: "The Silent Algorithm",
  author: "R. Menon",
  genre: "Technology",
  publishedYear: 2022,
  formats: ["Hardcover", "Digital", "Audiobook"],
  specs: { pages: 340, fileSize_MB: 4.2, drm: true, narrator: "K. Rao", runtime_min: 610 },
  avgRating: 4.6,
  reviews: [
    { member_id: "M1001", rating: 5, comment: "Excellent deep dive into digital systems design.", date: ISODate("2023-01-15") },
    { member_id: "M1042", rating: 4, comment: "Good but the Digital edition had formatting issues.", date: ISODate("2023-02-20") }
  ]
},
{
  title: "Ocean of Code",
  author: "A. Fernandes",
  genre: "Technology",
  publishedYear: 2021,

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},
{
  title: "Ocean of Code",
  author: "A. Fernandes",
  genre: "Technology",
  publishedYear: 2021,
  formats: ["Digital", "Paperback"],
  specs: { pages: 210, fileSize_MB: 2.1, drm: false },
  avgRating: 4.2,
  reviews: [
    { member_id: "M2001", rating: 4, comment: "Solid Digital read, clean layout.", date: ISODate("2022-05-10") },
    { member_id: "M2002", rating: 5, comment: "Best programming book this year.", date: ISODate("2022-06-01") }
  ]
},
{
  title: "Whispers of Yesterday",
  author: "S. Kapoor",
  genre: "Fiction",
  publishedYear: 2019,
  formats: ["Hardcover", "Paperback"],
  specs: { pages: 410, binding: "Sewn" },
  avgRating: 3.8,
  reviews: [
    { member_id: "M3001", rating: 3, comment: "Slow start but strong ending.", date: ISODate("2020-01-15") },
    { member_id: "M3002", rating: 4, comment: "Beautifully written prose.", date: ISODate("2020-02-11") }
  ]
},
{
  title: "Quantum Horizons",
  author: "R. Menon",
  genre: "Science",
  publishedYear: 2023,
  formats: ["Digital", "Audiobook"],
  specs: { fileSize_MB: 5.6, drm: true, narrator: "P. Iyer", runtime_min: 540 },
  avgRating: 4.9,
  reviews: [
    { member_id: "M4001", rating: 5, comment: "The Digital format made annotations so easy.", date: ISODate("2023-08-01") },
    { member_id: "M4002", rating: 5, comment: "Mind-bending and well narrated.", date: ISODate("2023-09-14") }
  ]
}
])
}
acknowledged: true,
insertedIds: {
  '0': ObjectId('6a89329a69041f57525a779f'),
  '1': ObjectId('6a89329a69041f57525a779a'),
  '2': ObjectId('6a89329a69041f57525a77a1'),
  '3': ObjectId('6a89329a69041f57525a77a2')
}
publishedYear: 2021,

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mongosh mongodb+srv://<credentials>@cluster0.eay9yl.mongodb.net/
}
Atlas atlas-4upy6x-shard-0 [primary] BookFlowDB> db.book_metadata.find({ avgRating: { $gt: 4 } }, { title: 1, avgRating: 1, _id: 0 })
db.book_metadata.find({ formats: { $in: ["Digital", "Audiobook"] } }, { title: 1, formats: 1, _id: 0 })
db.book_metadata.find({ avgRating: { $gt: 4 }, formats: { $in: ["Digital"] } })
db.book_metadata.find({ formats: { $regex: "Digital", $options: "i" } }, { title: 1, formats: 1 })
db.book_metadata.find({ "reviews.comment": { $regex: "Digital", $options: "i" } }, { title: 1, "reviews.$": 1 })
{
  _id: ObjectId('6a89329a69041f57525a779f'),
  title: 'The Silent Algorithm',
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: ISODate('2023-01-15T00:00:00.000Z')
    }
  ]
},
{
  _id: ObjectId('6a89329a69041f57525a77a0'),
  title: 'Ocean of Code',
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: ISODate('2022-05-10T00:00:00.000Z')
    }
  ]
},
{
  _id: ObjectId('6a89329a69041f57525a77a2'),
  title: 'Quantum Horizons',
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: ISODate('2023-08-01T00:00:00.000Z')
    }
  ]
}
Atlas atlas-4upy6x-shard-0 [primary] BookFlowDB> db.book_metadata.aggregate([
  { publishedYear: 2021,

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mongosh mongodb+srv://<credentials>@cluster0.eay9yl.mongodb.net/
{
  _id: ObjectId('6a89329a69041f57525a77a2'),
  title: 'Quantum Horizons',
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: ISODate('2023-08-01T00:00:00.000Z')
    }
  ]
}
Atlas atlas-4upy6x-shard-0 [primary] BookFlowDB> db.book_metadata.aggregate([
  { $match: { publishedYear: { $gt: 2020 } } },
  { $unwind: "$reviews" },
  { $group: {
    _id: "$title",
    avgReviewRating: { $avg: "$reviews.rating" },
    totalReviews: { $sum: 1 },
    author: { $first: "$author" }
  }},
  { $sort: { avgReviewRating: -1 } },
  { $project: {
    _id: 0,
    title: "$_id",
    author: 1,
    avgReviewRating: { $round: ["$avgReviewRating", 2] },
    totalReviews: 1
  }},
  { $group: {
    _id: 0,
    totalReviews: 2,
    author: 'R. Menon',
    title: 'Quantum Horizons',
    avgReviewRating: 5
  }},
  { $group: {
    _id: 0,
    totalReviews: 2,
    author: 'A. Fernandes',
    title: 'Ocean of Code',
    avgReviewRating: 4.5
  }},
  { $group: {
    _id: 0,
    totalReviews: 2,
    author: 'R. Menon',
    title: 'The Silent Algorithm',
    avgReviewRating: 4.5
  }},
  { publishedYear: 2021,

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mongosh mongodb+srv://<credentials>@cluster0.eay9jl.mongodb.net/
}
avgReviewRating: 4.5
}
Atlas atlas-4upj6x-shard-0 [primary] BookFlowDB> db.book_metadata.createIndex({ "reviews.comment": "text" })

db.book_metadata.find(
  { $text: { $search: "Digital" } },
  { title: 1, score: { $meta: "textScore" } }
).sort({ score: { $meta: "textScore" } })

db.book_metadata.find({ $text: { $search: "Digital" } }).explain("executionStats")

db.book_metadata.find({ "reviews.comment": { $regex: "Digital", $options: "i" } }).explain("executionStats")

db.book_metadata.createIndex({ formats: 1 })
db.book_metadata.createIndex({ avgRating: -1 })
db.book_metadata.find({ avgRating: { $gt: 4 } }).explain("executionStats")

{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlowDB.book_metadata',
    parsedQuery: { avgRating: { '$gt': 4 } },
    indexFilterSet: false,
    queryHash: '85B8A8276',
    planCacheKey: '6284A3BD',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExploreReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'FETCH',
      inputStage: {
        stage: 'IXSCAN',
        keyPattern: { avgRating: -1 },
        indexName: 'avgRating_-1',
        isMultikey: false,
        multikeyPaths: { avgRating: [] },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'forward',
        indexBounds: { avgRating: [ '[inf.0, 4)' ] ] }
      }
    },
    rejectedPlans: []
  },
  publishedYear: 2021,

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    isPartial: false,
    indexVersion: 2,
    direction: 'forward',
    indexBounds: { avgRating: [ '[inf.0, 4)' ] ] }
  }
},
rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 3,
  executionTimeMillis: 1,
  totalKeysExamined: 3,
  totalDocsExamined: 3,
  executionStages: {
    isCached: false,
    stage: 'FETCH',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    docsExamined: 3,
    alreadyHasObj: 0,
    inputStage: {
      stage: 'IXSCAN',
      nReturned: 3,
      executionTimeMillisEstimate: 0,
      works: 4,
      advanced: 3,
      needTime: 0,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      keyPattern: { avgRating: -1 },
      indexName: 'avgRating_-1',
      isMultikey: false,
      multikeyPaths: { avgRating: [] },
      isUnique: false,
      isSparse: false,
      isPartial: false,
      indexVersion: 2,
      direction: 'forward',
      indexBounds: { avgRating: [ '[inf.0, 4)' ] },
      keysExamined: 3,
      publishedYear: 2021,

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mongosh mongodb+srv://<credentials>@cluster0.eay9vj1.mongodb.net/
  multiKeyPaths: { avgRating: [] },
  isUnique: false,
  isSparse: false,
  isPartial: false,
  indexVersion: 2,
  direction: 'forward',
  indexBounds: { avgRating: [ '[inf,0, 4)' ] },
  keysExamined: 3,
  seeks: 1,
  dupsTested: 0,
  dupsDropped: 0
}
},
queryShapeHash: '17380C6D40919E08A14FE356779B269DD1F1431796C43D57B22D5D8FECABA3A0',
command: {
  find: 'book_metadata',
  filter: { avgRating: { '$gt': 4 } },
  $db: 'BookFlowDB'
},
serverInfo: {
  host: 'ac-pvuebaw-shard-00-02.eay9vj1.mongodb.net',
  port: 27017,
  version: '8.0.29',
  gitVersion: '559d67c651f7393e62757234a8b25cb7a8622148'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 33554432,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
  clusterTime: Timestamp({ t: 1787376305, i: 26 }),
  signature: {
    hash: Binary.createFromBase64('FTPb6Xdn1QxqcR3BMK1Hbru1Vnc=', 0),
    keyId: Long('7616446989839892482')
  }
},
operationTime: Timestamp({ t: 1787376305, i: 26 })
}
Atlas atlas-4upy6x-shard-0 [primary] BookFlowDB>
  publishedYear: 2021,
```